

Flood Ready: Flood Survival Game

An Educational Web-Based Game for Climate Disaster Preparedness

RMIT Hackathon 2025 - Challenge 3: Vibe Coding - Play to Impact

Team: AIMS

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Repository: https://github.com/Pandazelf/AIMS_Sample_Game_Project_Hackathon

1. Introduction

Flood Ready is an interactive, browser-based educational game designed to teach flood survival and emergency preparedness skills to Vietnamese communities, particularly those in the Mekong Delta region. The game simulates a realistic flood emergency scenario where players must make critical decisions about resource collection, emergency response, and survival strategies.

Built entirely with HTML, CSS, and JavaScript, the game runs completely in the browser without requiring any installation or backend infrastructure. Players experience three distinct phases: emergency preparation (60 seconds to collect essential items), crisis decision-making (10 randomly selected flood scenarios), and educational feedback (comprehensive results with real safety tips).

The game addresses a pressing social issue in Vietnam—increasingly severe and frequent flooding due to climate change—by providing an engaging, risk-free environment where players can learn life-saving skills through interactive gameplay.

2. Game Theme & Topic Justification

2.1 The Problem: Vietnam's Flood Crisis

Vietnam faces escalating flood disasters, particularly in the Mekong Delta and central regions. In 2024 alone, floods caused:

- **Billions of Vietnamese Dong** in economic damage
- **Displacement of thousands** of families from their homes
- **Loss of agricultural livelihoods** in rice-farming communities
- **Health crises** from contaminated water and disease outbreaks

- **Infrastructure destruction** affecting schools, hospitals, and essential services

The Mekong Delta, home to approximately **17 million people**, experiences annual flooding that is intensifying due to:

- **Climate change** increasing rainfall intensity
- **Rising sea levels** exacerbating coastal flooding
- **Deforestation** reducing natural water absorption
- **Urbanization** creating inadequate drainage systems

2.2 Why This Topic Matters

Knowledge Gap: Many Vietnamese families, especially in rural areas, lack comprehensive knowledge about:

- Proper emergency preparedness procedures
- Safe evacuation timing and protocols
- Water contamination dangers
- Resource management during crises
- Community solidarity importance

Traditional Education Limitations: Conventional flood safety education through pamphlets, posters, or lectures has limited engagement and retention. People learn best through experience, but real flood disasters are too dangerous for practice.

Gaming as Education: Interactive games provide a safe environment to:

- Experience realistic emergency scenarios
- Learn through consequences of decisions
- Repeat and improve performance
- Retain information through active engagement
- Build muscle memory for crisis response

2.3 Target Audience

- **Primary:** Vietnamese families in flood-prone regions (Mekong Delta, Central Vietnam)

- **Secondary:** Students learning disaster preparedness
 - **Tertiary:** Emergency response volunteers and community leaders
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3. Potential Impact

3.1 Educational Impact

Flood Ready teaches 20+ critical flood safety lessons including:

- **Emergency Preparedness:** What items to pack in go-bags, importance of documents and cash
- **Evacuation Procedures:** When to leave, never ignoring official warnings, helping vulnerable neighbors
- **Water Safety:** Never drinking or using floodwater, contamination risks, disease prevention
- **Medical Emergencies:** First aid for wounds, managing chronic conditions, recognizing symptoms
- **Electrical Hazards:** Avoiding downed power lines, generator safety, carbon monoxide risks
- **Resource Management:** Strategic use of limited supplies, sharing with community
- **Psychological Resilience:** Managing panic, combating rumors, maintaining hope

3.2 Behavioral Change

By experiencing consequences in a safe environment, players develop:

- **Automatic responses** to flood warnings (immediate evacuation)
- **Better judgment** about resource prioritization
- **Community-oriented thinking** (helping elderly, sharing supplies)
- **Skepticism toward rumors** (trusting official sources)
- **Preparedness habits** (maintaining emergency kits)

3.3 Community Impact

If widely adopted, the game could:

- **Reduce flood-related deaths** through better preparedness
- **Decrease disease outbreaks** by teaching water contamination risks
- **Strengthen community solidarity** through emphasis on collective survival
- **Improve emergency response** by educating future volunteers
- **Lower economic losses** through earlier evacuation and better resource management

3.4 Scalability

The game's design allows for:

- **Language expansion** (currently Vietnamese/English, can add more)
 - **Regional customization** (adapt scenarios for different flood-prone areas)
 - **Content updates** (add new scenarios based on recent disasters)
 - **Integration with education** (use in schools, community centers)
 - **Offline distribution** (works without internet once loaded)
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4. Technology Stack

4.1 Core Technologies

Frontend Framework:

- **HTML5** - Semantic structure, accessibility features
- **CSS3** - Advanced styling, animations, gradients, responsive design
- **Vanilla JavaScript (ES6+)** - Game logic, state management, DOM manipulation

Why No External Libraries?

To meet hackathon requirements of browser-only execution without installation, we used pure web technologies. This ensures:

- Zero dependencies
- Instant loading
- Works offline
- Maximum compatibility

- Minimal file size

4.2 AI Tools Used

GitHub Copilot Chat:

- **Game concept brainstorming** - Discussed simplified 3-phase game structure
- **Scenario generation** - Created 20 realistic flood scenarios based on Vietnam context
- **Code implementation** - Generated HTML/CSS/JavaScript for all game systems
- **Bug fixing** - Debugged item consumption, rating display, scenario logic
- **Refinement** - Polished UX, improved accessibility, optimized performance

ChatGPT/Claude (Conceptual):

- **Educational content validation** - Verified flood safety information accuracy
- **Vietnamese context research** - Ensured cultural and geographic relevance
- **Lesson formulation** - Crafted concise, actionable learning takeaways

Process:

All development was "vibe coded" using AI assistance. Every feature, from initial HTML structure to complex inventory management, was built through conversational prompting with AI coding assistants. This approach allowed rapid iteration and implementation of sophisticated game mechanics within hackathon time constraints.

4.3 Design & Assets

Visual Design:

- **CSS Gradients** - Water-themed blue gradients, emergency red warnings, success green
- **Unicode Emoji** - All icons (💧 🚒 🚨 📄 etc.) are built-in characters, no image files
- **Animations** - Pure CSS (pulse, shake, fade-in, hover effects)
- **Responsive Design** - Mobile-first approach with CSS Grid and Flexbox

No External Image Generation:

All visual elements are code-based (CSS + emoji), requiring no AI image generators like DALL-E or Midjourney.

4.4 Development Environment

- **Code Editor:** Visual Studio Code with Copilot extension
 - **Version Control:** Git & GitHub
 - **Testing:** Modern browsers (Chrome, Firefox, Safari, Edge)
 - **Deployment:** GitHub Pages (static hosting)
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5. Overview of Game Mechanics

5.1 Three-Phase Gameplay

Phase 1: Emergency Preparation (60 seconds)

Players face an urgent flood warning and must quickly collect **8 essential items** from their house. The game presents **14 available items**:

Consumable Items (used once):

- Bottled Water 💧
- Food 🍲
- First Aid Kit 🩹
- Cash Money 💵
- Medicine 💊
- Plastic Bags 🛍️

Non-Consumable Items (reusable):

- Flashlight 🔦
- Important Documents 📄
- Phone 📱
- Radio 📻
- Rope 🪄
- Whistle 📯

- Blanket 🛏️
- Matches 🔥

Mechanics:

- Real-time countdown creates urgency
- Click-to-collect interface
- Visual feedback (items turn green when collected)
- Backpack capacity limit forces prioritization
- Strategic choice: versatile items vs. specific-use items

Phase 2: Flood Crisis (10 scenarios)

Players face **10 randomly selected scenarios** from a pool of **20 diverse situations**, including:

Categories:

- Evacuation & Warnings (immediate decisions, transportation, night evacuation)
- Water Safety (drinking, cooking, wound care, baby formula)
- Medical Emergencies (chronic medication, fever, diabetic shock)
- Rescue & Signaling (helicopter rescue, boat ethics)
- Communication (rumor control, information verification)
- Shelter Life (food distribution, cold weather, generator safety)
- Psychological Challenges (panic attacks, property vs. safety)

Mechanics:

- Each scenario presents 3-4 choices
- Choices are **dynamically available** based on collected items
- Missing required items **disables** certain choices (grayed out with "❌ Missing: [item]")
- At least **2 choices always available** (no-item fallbacks)
- Consumable items are **removed from inventory** after use

- Non-consumable items remain available
- Immediate feedback: Safe  / Risky  / Danger 
- Each choice includes an **educational lesson**

Example Scenario:

"Thirst Crisis at Shelter - You've been at evacuation center for 8 hours. You're very thirsty. Water distribution ran out."

- **Choice A:** Drink bottled water (requires  , consumes ) → **SAFE**
- **Choice B:** Share water with sick child (requires  , consumes ) → **SAFE**
- **Choice C:** Wait for officials (no items) → **RISKY**
- **Choice D:** Drink floodwater (no items) → **DANGER**

Phase 3: Results & Education

5-Star Rating System:

-  **Excellent (10/10 safe)** - Perfect preparation and decision-making
-  **Good (7-9/10 safe)** - Strong flood safety knowledge
-  **Average (5-6/10 safe)** - Basic understanding, needs practice
-  **Poor (3-4/10 safe)** - Many dangerous mistakes
-  **Disaster (0-2/10 safe)** - Life-threatening decisions

Feedback Provided:

- Detailed score breakdown (safe/risky/danger counts)
- Remaining inventory (unused items)
- Comprehensive lessons learned (from choices made)
- Real emergency contact information (Vietnam: 113 emergency, 115 health)
- Flood safety rules checklist
- Replay option to improve score

5.2 Core Game Systems

Inventory Management System:

JavaScript

- Items have "helps" array (e.g., water helps: ["water", "thirst", "drink"])
- Choices have "needs" array (e.g., needs: ["water", "medical"])
- Choices have "consumes" array (what gets used up)
- Real-time inventory updates after each decision
- Visual indicators show what will be consumed

Random Scenario Selection:

JavaScript

- Shuffle all 20 scenarios
- Select first 10 for current playthrough
- Ensures different experience each game
- Millions of possible combinations ($20! / (20-10)! = 670,442,572,800$)

Dynamic Choice Availability:

JavaScript

- Check player inventory for required item capabilities
- Disable choices if items missing
- Show clear "Missing: [item]" feedback
- Ensure at least 2 choices remain available
- No possibility of getting stuck

Educational Feedback Loop:

JavaScript

- Immediate consequence shown after each choice
- Lesson saved and displayed in results
- Unique lessons prevent repetition
- Real-world context in every scenario

5.3 Replayability Features

- **Random scenarios** ensure different games every time
 - **5-star rating** provides clear improvement goal
 - **Item strategy** can be optimized across playthroughs
 - **Educational depth** - new lessons discovered each game
 - **Challenge modes** (theoretical future addition: Easy/Normal/Hard difficulty)
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6. Reflection

6.1 Development Process

What Went Well:

AI-Assisted Development:

Using GitHub Copilot and ChatGPT as pair programming partners was transformative. The ability to describe game mechanics in natural language and receive working code dramatically accelerated development. Complex systems like dynamic inventory management and random scenario selection were implemented in hours rather than days.

Iterative Refinement:

The project evolved significantly from initial concept (simple card game) to final product (sophisticated resource management game). Each iteration added depth: 5→20 scenarios, 2→4 choices, 3→5 star rating, static→dynamic inventory. This organic growth resulted in a much more engaging final product.

Educational Effectiveness:

Balancing entertainment and education was achieved through consequence-based learning. Players don't feel lectured—they discover why certain actions are dangerous through experiencing outcomes. The immediate feedback loop reinforces learning effectively.

Challenges Faced:

Game Balance:

Initial versions had scenarios assuming players possessed certain items, leading to unwinnable situations. Solution: Redesigned every scenario to include at least two no-item-required choices, ensuring playability regardless of preparation phase performance while still rewarding good item selection.

Item Consumption Logic:

Implementing the consumable/non-consumable system required careful tracking of inventory state across scenarios. Debugging involved extensive testing to ensure items were removed correctly, choices updated dynamically, and the UI reflected real-time changes.

Scenario Quality:

Creating 20 realistic, educational scenarios required balancing:

- Realism (based on actual Vietnam flood situations)
- Variety (covering different emergency aspects)
- Fairness (appropriate difficulty distribution)
- Educational value (each teaching distinct lessons)

Research into Vietnamese Red Cross guidelines and recent flood reports ensured authenticity.

Mobile Responsiveness:

Ensuring the game worked well on phones required careful CSS design. The solution involved CSS Grid with responsive breakpoints, touch-friendly button sizing, and readable text scaling across devices.

6.2 Learning Outcomes

Technical Skills:

- **Advanced JavaScript:** State management, array manipulation, dynamic DOM updates
- **CSS Mastery:** Gradients, animations, responsive design, flexbox/grid systems
- **Game Design:** Balance, feedback loops, progression systems, UX optimization
- **AI Collaboration:** Effective prompting, iterative refinement, debugging with AI assistance

Domain Knowledge:

- **Disaster Preparedness:** Deep understanding of flood safety protocols
- **Vietnamese Context:** Mekong Delta geography, cultural considerations, local challenges
- **Educational Design:** How to teach through gameplay, consequence-based learning

- **Emergency Response:** Medical emergencies, evacuation procedures, community resilience

Soft Skills:

- **Problem Solving:** Debugging complex logic errors, balancing competing design goals
- **User Empathy:** Designing for diverse audiences, accessibility considerations
- **Time Management:** Prioritizing features within hackathon constraints
- **Communication:** Writing clear educational content, user feedback design

6.3 If I Had More Time

Expanded Features:

- **Multiplayer Mode:** Cooperative survival with resource sharing decisions
- **Career Mode:** Progress through multiple flood seasons, building long-term preparedness
- **Regional Variants:** Customized scenarios for different Vietnamese provinces
- **Difficulty Levels:** Easy (more items, simpler scenarios), Hard (fewer items, complex dilemmas)
- **Achievement System:** Badges for perfect runs, helping others, discovering all scenarios
- **Sound Design:** Background ambience, notification sounds, feedback audio
- **Accessibility:** Screen reader support, colorblind modes, keyboard-only navigation

Community Features:

- **Leaderboards:** Compare scores with friends and community
- **Scenario Sharing:** Community-created scenarios reviewed by educators
- **Translation Contributions:** Crowdsourced translations to other languages
- **Educational Integration:** Teacher dashboard, classroom tracking, lesson plans

Technical Improvements:

- **Backend Integration:** Save progress, track learning outcomes over time

- **Analytics:** Understand which scenarios are most educational, where players struggle
- **A/B Testing:** Optimize scenario wording, choice presentation for maximum impact
- **Performance Optimization:** Even faster loading, smoother animations

Partnerships:

- **Vietnamese Red Cross:** Official endorsement, distribution through their networks
 - **Schools & Universities:** Curriculum integration, student competitions
 - **Government Emergency Services:** Official disaster preparedness platform
 - **NGOs & Aid Organizations:** Training tool for volunteers and communities
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7. Conclusion

Flood Ready demonstrates how interactive gaming can effectively teach life-saving emergency preparedness skills. By combining realistic scenarios, strategic resource management, and immediate educational feedback, the game provides an engaging learning experience that traditional methods cannot match.

The project successfully addresses a critical social issue—flood disaster preparedness in Vietnam—through innovative use of web technologies and AI-assisted development. With 20 diverse scenarios, dynamic inventory management, and a 5-star rating system, the game offers significant replay value while ensuring players learn essential flood safety lessons.

Most importantly, **Flood Ready** has the potential to save lives. If adopted by Vietnamese communities, schools, and emergency services, the game could significantly improve flood preparedness, reduce casualties, and strengthen community resilience in the face of increasingly severe climate disasters.

The development process showcased the power of AI-assisted "vibe coding"—transforming ideas into functional, polished software through natural language conversation with AI coding assistants. This approach enabled the creation of a sophisticated educational game within hackathon time constraints, demonstrating the future of rapid software development.

Flood Ready is more than a game—it's a tool for community empowerment, disaster preparedness, and climate adaptation education.

Play Flood Ready and Learn to Save Lives

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RMIT Hackathon 2025 - Challenge 3 Submission

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